Thakur College of Engineering & Technology

HACKANOVA 5.0

Break The Loop, Create the future

PPT Template

Team Member Details

Team Name	**Mtx.Vanitra**			
Team Members >>	1 (Leader)	2	3	4
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Problem Statement

Globally, 1.5 billion experience hearing loss. In India, ~63 million have hearing impairments . Deaf-mute people often struggle to communicate with the hearing world due to a lack of interpreters, high costs, and technology dependence.

Communication barriers isolate 63M+ deaf-mute individuals in India.

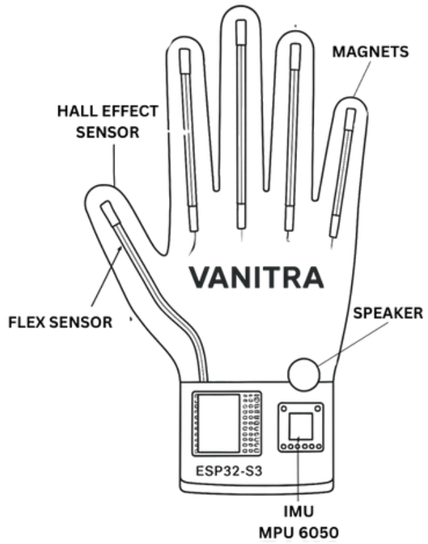
- **Sign language gloves only translate ONE WAY (gesture → speech)**
- **No portable systems for receiving speech as tactile/text feedback**
- **Existing solutions need smartphones/internet → not truly accessible**
- **Vanitra solves this with offline bidirectional wearable communication**

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Working Principle

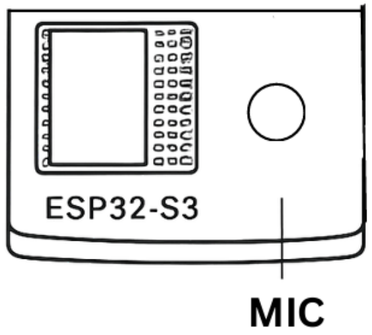
Glove: Flex + Hall + IMU sensors → ESP32-S3 → Embedded ML → SPEECH output

Smart glove uses 5 flex sensors (GPIO 4-7) for finger bends, hall sensors (GPIO 9-13) for contacts, and MPU6050 IMU (I2C 16/17) for orientation. 100Hz data → TinyML classifier on ESP32-S3 → 90% ASL accuracy → speaker output in <500ms.



Wristband: Microphone → Speech recognition → Text

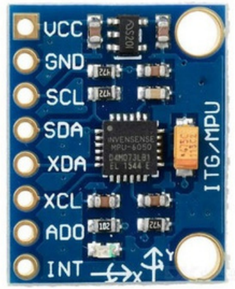
Electret mic captures speech → keyword detection → Morse encoder (50ms dots, 150ms dashes) → PWM haptic motor vibrations for silent message reception. Fully offline, no phone needed.



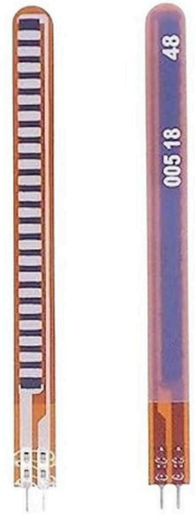
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Required Components

- **ESP32-S3 microcontroller**
- **5x Flex sensors**
- **Hall effect sensors + magnets**
- **MPU6050 IMU**
- **Microphone module**
- **Haptic motor + Speaker**
- **Li-ion battery + regulator**



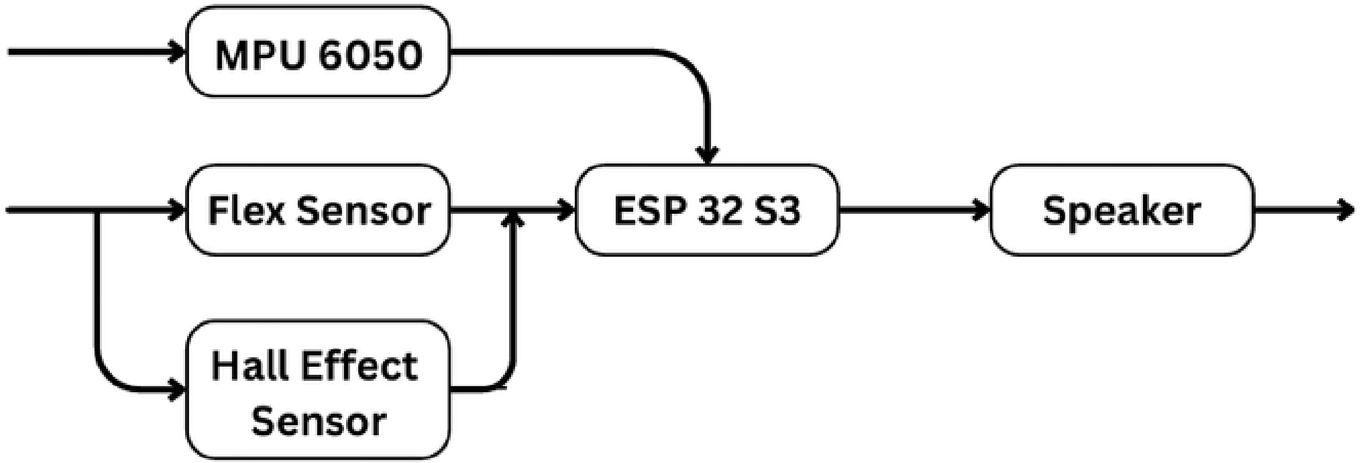




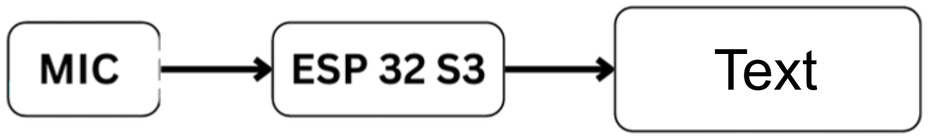
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Flowchart/Block Diagram/ Workflow Diagram

Gesture Input →



Speech Input →



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Expected Outcomes

- **90%+ gesture recognition accuracy (trained ASL alphabet + phrases)**
- **Real-time bidirectional communication (<500ms latency)**
- **Offline operation (no phone/internet needed)**
- **Portable wearable (glove + wristband)**
- **Social impact: Empowers deaf-mute independence and inclusion**

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Execution Plan

Hour 0-6: Hardware Bring-up & Sensor Calibration

- Flex sensor voltage divider setup (ADC GPIO4-7)
- Hall sensor debouncing (GPIO9-13)
- MPU6050 I2C init (GPIO16 SDA, GPIO17 SCL)
- Mic preamp + haptic PWM driver
- Per-user flex min/max calibration routine

Hour 6-18: Embedded ML Pipeline & Firmware text

- TinyML gesture classifier deployment (quantized model)
- Feature extraction: flex_norm + hall_bits + IMU_deltas
- Morse encoder with precise dot/dash timing (50ms/150ms)
- Non-blocking state machine: IDLE→SENSE→CLASSIFY→OUTPUT
- Serial logging + confidence gating (>0.85 threshold)

Hour 18-28: Integration & Performance Tuning

- End-to-end latency profiling (<500ms target)
- Gesture→speech pipeline validation
- Speech→haptic Morse rendering/text on display (legibility test)
- Battery duty cycling + power profiling
- Error handling (sensor timeout, low confidence fallback)

Hour 28-36: Demo Prep & Stress Testing

- 20-gesture accuracy validation (>90%)
- Multi-user calibration demo
- Real conversation showcase (bidirectional flow)
- Final prototype packaging + video recording
- Judge Q&A technical rehearsal

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Potential Problems and Limitations

Technical Challenges:

- **Sensor drift: Flex baselines shift**
- **IMU noise: Vibration artifacts**
- **ML inference latency: >500ms**
- **Hall sensor false triggers: Magnetic interference**
- **Mic noise floor: Ambient pickup**

Hardware Constraints:

- **GPIO conflicts: Flex vs touch pins**
- **Power spikes: Haptic + speaker**
- **Wiring strain: Glove movement**

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